1. A box contains 4 blue and 2 red balls. We take one ball out of it, note down its color and put it back in the box. Thereafter we take out another ball and find it to be blue. Let the probability that the first ball was red given that the second ball is blue be **p**.
   Suppose instead of putting back the first ball in the box we threw it away. Now let the probability that the first ball was red given that the second ball is blue be **q**. What is the ratio of **q/p**. Round off your answer till 1 decimal point. (e.g. if your answer is 3.14159… then write 3.1)
   Ans. 1.2

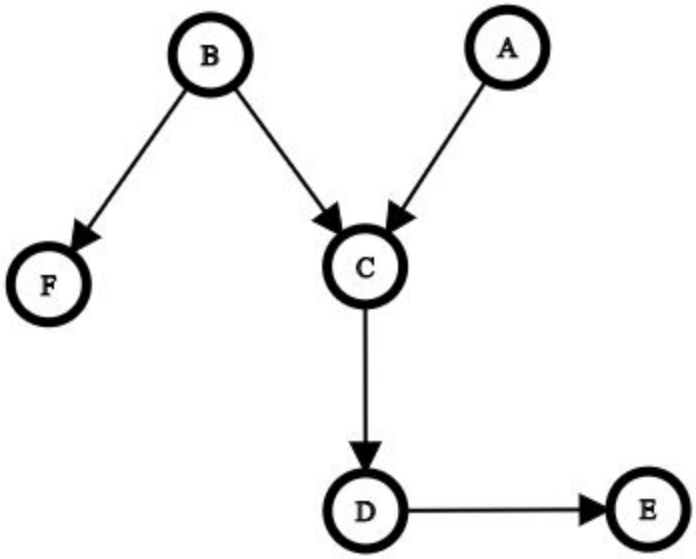
   When we sample with replacement, any two samples are independent of each other. So, $P(Ball_1=Red|Ball_2=Blue) = P(Ball_1=Red) = P(Ball=Red) = 2/6$

   When we sample without replacement, then the two samples are not independent and we will need Bayes Rule here.

   $P(Ball_1=Red|Ball_2=Blue) = P(Ball_2=Blue|Ball_1=Red).P(Ball_1=Red)/ P(Ball_2=Blue)$
   $= (4/5 * 2/6) / (4/5 * 2/6 + 3/5 * 4/6) = 8/20$

   q/p = (8/20)/(2/6) = 1.2

2. Consider the following Bayesian Network. Given evidence about C which of the following pair of variables are conditionally independent.



a. A and B
b. A and D
c. D and E
d. A and F

A and B are not conditionally independent given C since it is a converging connection and information can flow between A and B if and only if we do have evidence at C or any descendent of C (such as D or E).

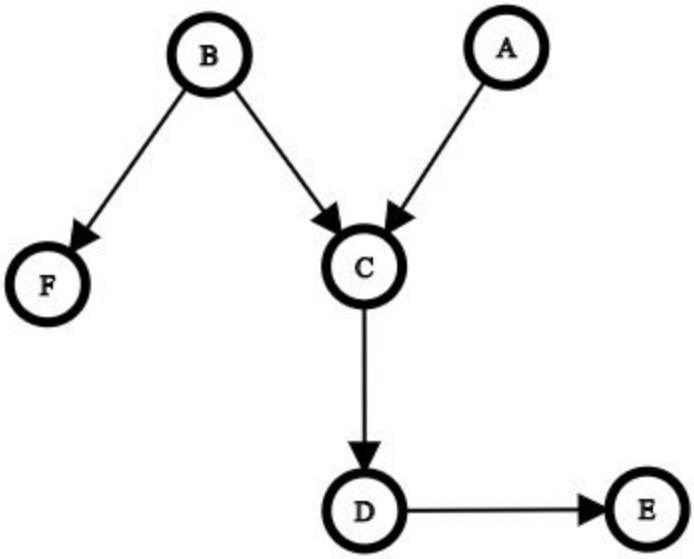
A and D are conditionally independent given C since it is a linear connection and information can flow between A and D if and only if we do not have evidence at C.

D and E are directly connected so they cannot be conditionally independent given C.

A and F are not conditionally independent since A and B are not conditionally independent and thus information can flow between A and F via B.

See lecture slides http://www.cse.iitd.ac.in/~mausam/courses/col333/autumn2019/lectures/10-bayesnets.pdf for further explanation.

3. Consider the following Bayesian Network. Which of the following variables belong to Markov Blanket of F.



a. A
b. B
c. C
d. D

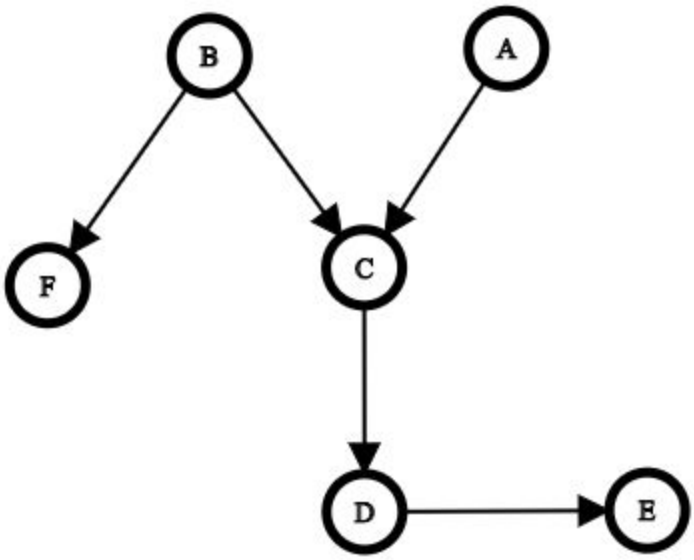
Markov Blanket of X = Par(X) ∪Childs(X) ∪ Par(Childs(X))
Par(F) = {B}
Child(F) = {}
Par(Child(F)) = {}
Markov Blanket of F = {B}

4. Suppose A and B are conditionally independent given C. Then which of the following are true:
    a. P(A,B) = P(A).P(B)
    b. P(A,B,C) = P(A|B,C).P(B|C).P(C)
    c. P(A,B,C) = P(A|C).P(B|C).P(C)
    d. P(A,B|C) = P(A|C).P(B|C)

P(A,B,C) = P(A|B,C).P(B|C).P(C) (factorization of joint probability)
= P(A|C).P(B|C).P(C) (conditional independence of A and B)

P(A,B|C) = P(A|B,C).P(B|C) (factorization of conditional probability)
= P(A|C).P(B|C) (conditional independence of A and B)

5. How many minimum parameters are required to represent the following Bayesian Network, if each variable is Boolean?



Ans. 12

P(B), P(A) 1 parameter each
P(F|B) - 2 parameters
P(C|A,B) - 4 parameters
P(D|C) - 2 parameters
P(E|D) - 2 parameters

6. There are two boxes A and B. Box A contains n red balls and 2 blue balls while box B contains 2 red balls and n blue balls. One of the two boxes is selected at random and two balls are drawn from it without replacement. If both the balls drawn are red, the probability of box A being selected is 6/7. Find the value of n.
   Ans: 4

   Let C be the event that the two balls drawn were red.

   $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$

   $P(C \mid A) = \frac{n.(n-1)}{(n+2).(n+1)}$, $P(C \mid B) = \frac{2.1}{(n+2).(n+1)}$

   Using Bayes theorem, we have,

   $P(A \mid C) = \frac{P(C \mid A).P(A)}{P(C \mid A).P(A) + P(C \mid B).P(B)}$

   i.e. $\frac{n.(n-1)}{n.(n-1)+2} = \frac{6}{7}$

   i.e $7.n.(n-1) = 6.n.(n-1) + 12$

   i.e. $n.(n-1) = 12$

   i.e. $n = 4$

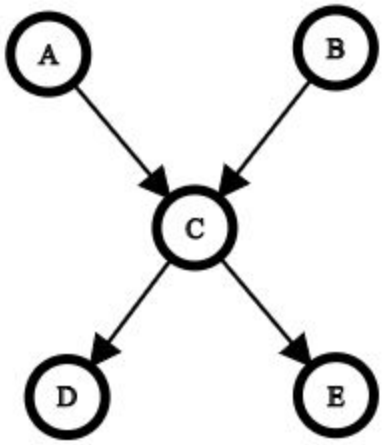
7. Find the difference in the minimum number of parameters required to specify a Bayesian network of 6 variables when there are no independencies and when all the variables are mutually independent.
   Ans: 57

   Consider n variables in general.
   When there are no independencies, we need to have every combination of variables as a parameter except for one i.e. $(2^n - 1)$ parameters. When all the variables are mutually independent we just need one value $P(x_i)$ for each variable x_i i.e. n parameters.
   Difference = 2^n - n - 1. For n = 6, the difference is 57.

8. Which of the following statements is/are correct?
   a. **P(a | b, c) + P(¬a | b, c) = 1**
   b. P(a | b, c) + P(a | ¬b, ¬c) = 1
   c. **P(a | b, ¬c) + P(a | b, c) = P(a | b)**
   d. **P(a, b | c) = P(a | c).P(b | a, c)**

Option A is correct because the sum of probabilities of occurrence of an event X and not occurence of the event X is 1 given any evidence. Option B is incorrect because it is not supported by any result. In fact, we need to consider the cases when one of B and C occurs and the other does not. Then too, the LHS would sum to P(a) not 1. Option C is correct because the probability of any event can be split as the sum of probabilities of occurrence of that event given an event X and given the event not X. Option D is correct as it is the definition of conditional probability.

9. Consider the following Bayesian Network with the given information. What is the value of P(¬a | d, e) up to 3 decimal places?



P(a) = 0.001, P(b) = 0.002
P(c | a, b) = 0.95, P(c | a, ¬b) = 0.94, P(c | ¬a, b) = 0.29, P(c | ¬a, ¬b) = 0.001
P(d | c) = 0.9, P(d | ¬c) = 0.05
P(e | c) = 0.7, P(e | ¬c) = 0.01

Ans: 0.716

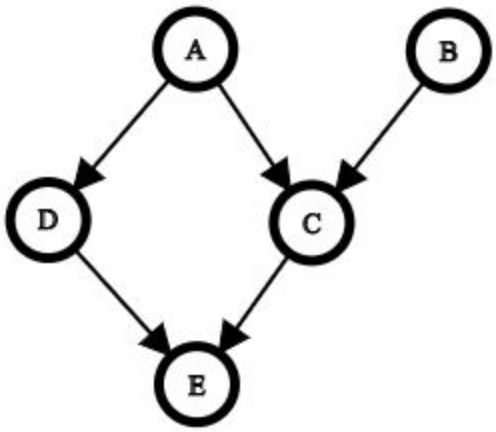
$$P(A,\ B,\ C,\ D, E)\ =\ P(A).P(B).P(C\mid A, B).P(D\mid C).P(E\mid C)$$

$$P(a,\ d,\ e)\ =\ \sum_{C}\sum_{B} P(a).P(B).P(C\mid a, B).P(d\mid C).P(e\mid C)\ =\ 0.00059224259$$

$$P(\neg a,\ d,\ e)\ =\ \sum_{C}\sum_{B} P(\neg a).P(B).P(C\mid \neg a, B).P(d\mid C).P(e\mid C)\ =0.001491857649$$

$$P(\neg a\mid d,\ e)\ =\ \frac{P(\neg a, d, e)}{P(d, e)}\ =\ \frac{P(\neg a, d, e)}{P(a, d, e) + P(\neg a, d, e)}\ =\ \frac{0.001491857649}{0.0005922459 + 0.001491857649}\ \simeq 0.716$$

10. Consider the following Bayesian Network.



Statement 1: B is independent of E given evidence at C.
Statement 2: B is independent of A given evidence at E.

a. Both statements are correct.
b. Both statements are incorrect.
c. Statement 1 is correct and Statement 2 is incorrect.
d. Statement 1 is incorrect and Statement 1 is correct.

Statement 1: Information can flow between E and A since we do not have evidence at D (linear connection). Information can flow between A and B since we have evidence at C (converging connection). Therefore, information can flow between E and B and B is not independent of C. Hence, the statement is false.

Statement 2: Information can flow between A and B since we have evidence at E which is a descendant of C (converging connection). Therefore, b is not independent of A. Hence, the statemtent is false.